

Potential of Wrist-worn Vibrotactile Feedback to Enhance the Perception of Virtual Objects during Mid-air Gestures

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ABSTRACT

Numerous systems based on mid-air gestures have recently been proposed as a digital variant of object manipulation with hands. Simultaneously, however, direct haptic feedback is lost, eliminating an important aspect that we are familiar with from real-life interaction. We believe that smartwatches, as widely used personal devices, could provide a platform for accessible, flexible, and unobtrusive integration of haptic feedback into mid-air gesture interaction. We prototyped a vibrotactile wrist band with four vibration actuators aiming at communicating invisible, undetectable virtual object properties such as electricity, weight, and tension into 3D haptic experiences. In this paper, we present findings from a user study ($n=18$) that examined the suitability of different vibration patterns (variation in intensity, temporal profile, rhythm, and location). Results show that all feedback variants have a positive impact on user experience (UX) when interacting with virtual objects. Constant, continuous patterns outweigh the other variants examined.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**; **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

Vibrotactile feedback, Usability/UX studies, Haptic perception, Multimodal interaction.

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1 INTRODUCTION

Natural user interfaces (NUIs) and especially mid-air gestures are increasingly used in human-computer interaction as technologies like Microsoft's Kinect or HoloLens become commercially available

and are technically mature. While the interaction with such technologies is natural and intuitive, they lack the problem of missing physical contact, i.e., direct haptic feedback. This limits perception in many practical cases, for example, when we want to know whether a (virtual) cooking plate is still hot, an electric fence is charged, or how heavy a vase is? In these cases, in reality, we would touch or pick up the object in question to get such information. But how do we touch or grasp something that is physically not present? When interacting with mid-air gestures, this is exactly the problem that arises. On the other hand, digital companions for daily life, such as smartwatches, are becoming increasingly popular and are always accessible. Smartwatch sales statistics show that only in 2021, 110 million people, representing 1.4% of the world population, will buy a smartwatch [24]. Smartwatches, via pairing with a smartphone, can already display smartphone information and generate fine-grained vibrotactile feedback using built-in vibration motors.

Based on the problem of a) not experiencing direct haptic feedback while performing mid-air gestures and b) the potential of utilizing smartwatches as accessible, flexible, and unobtrusive vibrotactile displays, we envision vibrotactile feedback around the wrist to convey invisible object properties of virtual environments. Especially physical forces or phenomena originating from objects are frequently occurring properties that are omnipresent through real interactions with objects. However, when performing gestures in the air, these effects are only indirect or not visible at all. To overcome this problem, we prototyped a vibrotactile wrist band and experimented with different vibration patterns to simulate a) physical forces (gravity and tension), b) phenomena like electricity, and c) spatial relation of those effects (i.e., distance, direction/orientation). The wearable can convey 3D haptic information around the wrist using four vibration actuators (on top, bottom, left, and right sides). Thereby, the vibrotactile feedback is not intended to replace the human's dominant visual channel, but rather to serve as a support to perceive object characteristics that are invisible to the eye, following a multimodal approach [25]. Subsequently, we present results from a user study comparing six vibration patterns differing in vibration intensity, temporal profile, rhythm, and position/location, based on the user's interaction (i.e., mid-air gestures or hand movement). Initial results show a high general acceptance/UX of the wrist band's additionally provided haptic feedback. Furthermore, we point out detailed benefits and limitations of parameters to design vibration patterns and conclude with limitations and ideas for ongoing research.

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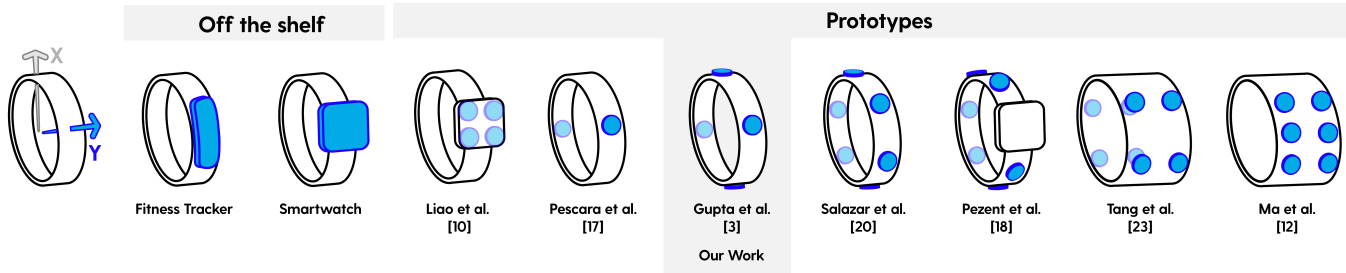


Figure 1: Overview of actuator alignment possibilities around the wrist.

2 RELATED WORK

As a physically and visually accessible body part, the wrist seems a promising area to attach vibrotactile displays. Research by Cholewiak et al. [2] shows that stimuli near the elbow and wrist – as the places near joints as physical reference points – achieve a higher detection rate of vibrotactile stimuli. Therefore, existing approaches demonstrate a wide range of different actuator placement possibilities to enable vibrations on the wrist, as shown in Figure 1.

Already today, off the shelf fitness tracker and smartwatches, such as the Fitbit Charge 4 or Apple Watch Series 5, can present information via vibrotactile stimuli, mostly as simple notifications. However, design possibilities are low due to the restriction to only one controllable actuator and corporate identity (CI) guidelines. Research has, therefore, proposed the use of multiple actuators to improve information value and transfer. Liao et al. [10], for example, utilized the housing bottom of a smartwatch-prototype to integrate four vibration motors – arranged as a 2x2 array with a distance of 3 cm to each other – and thus stimulate four areas of the dorsal side of the wrist. This placement enables the researchers to display alphanumeric letters by vibration. Several sides of the wrist can also be stimulated, as proposed by Pescara et al. [17]. Thereby, the display communicates vertical speeds for paragliders through two actuators placed dorsally and ventrally on the wrist. To also cover the horizontal position designation, Gupta et al. [3] created a continuous one-dimensional tactile display consisting of four actuators in combination with the phantom sensation. This way, the authors demonstrate its use as a tactile pointer for performing pointing, selection, and drag and drop tasks in two-dimensional space. A similar setup was followed by Salazar et al. [20] using a construct of six actuators around the wrist in combination with a phantom sense to provide vibrotactile cues for guidance through two-dimensional space. In the work of Pezent et al. [18], six vibration motors around the wrist supplement a squeezing mechanism to provide haptic feedback for a wide application area for the interaction with the virtual world, including surface and texture perception, simulating object weight and inertia, rendering force fields, and simulating reacting forces between hands for bimanual interactions [19]. To provide vibrotactile cues for navigation in three-dimensional space, Tang et al. [23] have chosen a cube-like arrangement of 8 actuators around the wrist to give three-dimensional feedback over 28 directions. Ma et al. [12] introduced a vibrotactile display on the wrist, constructed out of a 2x3 actuator array with a distance of 35mm to each other, to warn and guide drivers in automated vehicles about occurrences on the road. Based on the presented

related wrist-worn vibrotactile displays, we can conclude that their design and characteristics strongly depend on the tasks to be performed and the dimension in which the interaction takes place. For example, it makes sense that a three-dimensional arrangement of vibration motors is chosen for three-dimensional interactions. To save space and maintain the form factor of smartwatches, we use a one-dimensional array of vibration motors and present dimensional information through dynamically changing vibration patterns.

3 CONCEPT OF THE VIBROTACTILE WEARABLE

For our prototype, we integrated four horizontally arranged ERM actuators by NFP-Motor Co., Ltd. as an endless, one-dimensional vibrotactile display around the wrist (Figure 2). This arrangement allows to exceed the two-point discrimination threshold for both successive (240 mm) and continuous (180 mm) stimuli on the forearm [13], providing clear spatial cues and increasing the information content as suggested by [2, 14, 16]. The space between the actuators is in the effective range of 40-80 mm to apply the funneling illusion between the actuators as advised by [1]. Therefore, this arrangement enables us to provide spatial hints by adjusting the vibrotactile stimulus' position in a continuous sensation along the wrist if the hand's pose and orientation changes.

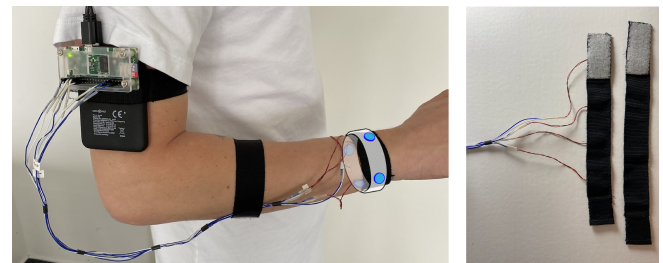


Figure 2: Left: A user wearing the prototype (i.e., power supply and controller mounted on the upper arm, four tactile actuators integrated into a wrist band). The strap on the forearm serves to fasten the cables. Right: Two linen wrist bands with integrated ERM actuators.

The used actuators reach frequencies of up to 250 Hz, which meet the skin's maximum sensitivity range towards vibrotactile stimuli [11]. A Raspberry Pi Zero W is used to control the actuators, defining their action potential via PWM and connected to a

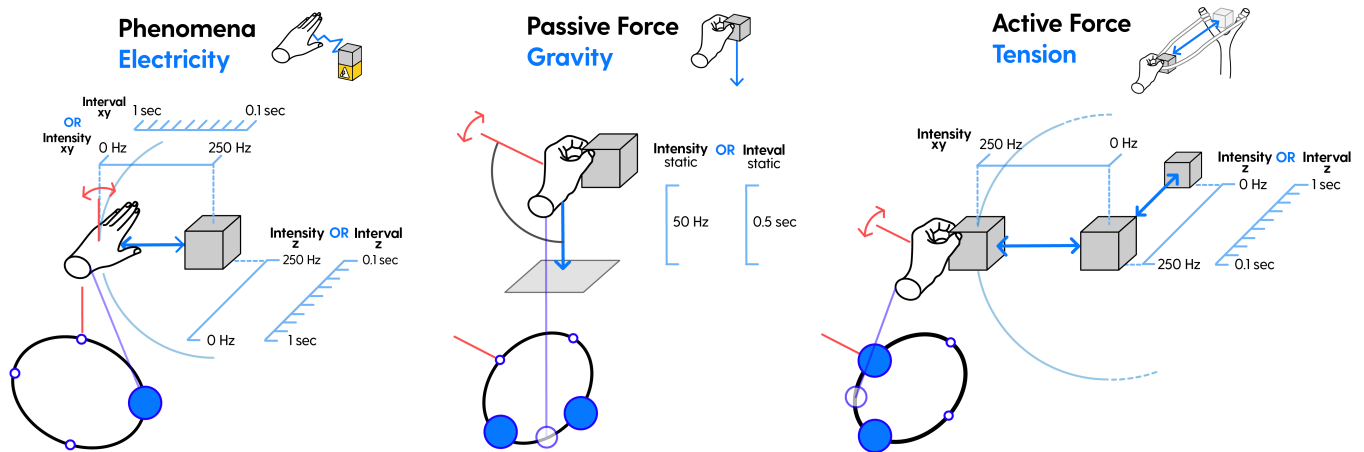


Figure 3: Exemplary behavior of vibrotactile feedback with individual actuator configuration (=blue dots) based on user’s hand position and orientation to an interactable object. Actuator size indicates the action potential (direction and magnitude of the effect).

battery (capacity 5000 mAh, 3.7 V). Counteracting the lack of flexibility towards humans’ different wrist circumferences, two wrist bands were built (140-200 mm and 190-250 mm) based on men and women’s average arm circumferences [5, 22]. The wristband also has continuous sewn-in 15mm pockets in which the motors can be placed, allowing a custom configuration for each user. Regarding the material, early tests showed that elastic material transmits vibrations between nearby actuators and leads to reduced spatial accuracy of tactile stimuli, so we chose pure linen for the wristband. Finally, due to the additional weight, we decided to mount the controller and power supply on the user’s upper arm to prevent arm fatigue.

3.1 Implementation of the Interaction

To test the display’s potential and develop the vibration patterns, we implemented an Unity3D application in which the user can interact with objects to experience different invisible object properties. User input is captured by a Leap Motion controller, which senses the user hand’s position, orientation, and gesture. This data is applied to a virtual representation of the hand per frame in the scene and displayed on a monitor. Simultaneously, all data for the vibrotactile messages are calculated by the application and forwarded to the display’s controller, transforming them into vibrotactile messages. The wireless data exchange happens in run time via WebSocket protocol to ensure a timely accurate representation of the vibrotactile feedback based on the user’s input.

3.2 Vibrotactile Feedback

To explore the numerous possibilities of shaping vibrotactile stimuli from multiple parameters, we created a modular construction kit as a basis for developing vibrotactile messages for a wrist-worn wearable. The components from which the patterns can be generated are listed below and can change dynamically based on the user’s hand position and orientation (Figure 3). In addition, Figure 4 provides an overview of how these components were combined to create

six different vibration patterns to represent the various use cases. For the selection of properties we have divided invisible object properties into three categories. For each one, a representative was selected for the study (section 4). **Phenomena:** Effects originating from objects that have no direct effect on its behavior, but can be naturally perceived through receptors of the skin (e.g. electricity). **Passive Force:** Forces that derive from the properties of an object and remain constant despite the user’s action (e.g. gravity). **Active Force:** Forces that originate from properties of the objects that change as a result of active interaction (e.g. tension). The choice is also based on the fact that these effects provide feedback on the status of interaction with an object.

3.2.1 Actuator configuration. We used two configurations (A4, I; see below) to deliver vibrotactile stimuli to the wrist to explore the potential of conventional wrist wearables and the possibilities of extending them to include spatial stimuli.

- **All Four (A4):** The vibrotactile pattern is transmitted synchronously by all four actuators. However, this configuration makes it impossible to provide spatial information around the wrist.
- **Individual (I):** Each actuator is assigned an individual intensity of the vibrotactile pattern depending on the user’s hand orientation. This configuration allows creating seamless spatial cues around the wrist about the forces’ direction by applying the funneling illusion between the actuators.

3.2.2 Tacton components.

- **Intensity (IY):** Depending on the distance and orientation of the user’s hand to a reference point, the intensity can be continuously varied between 0 and 250 Hz by changing the frequency of the actuators.
- **Interval (IL):** Pulse width modulation (PWM) was used to realize a temporal profile. The duration of the on-pulse was set to 0.1 sec, the duration of the interstimulus interval is varied in 0.1-second steps from 0.1 sec to a maximum of 1.0

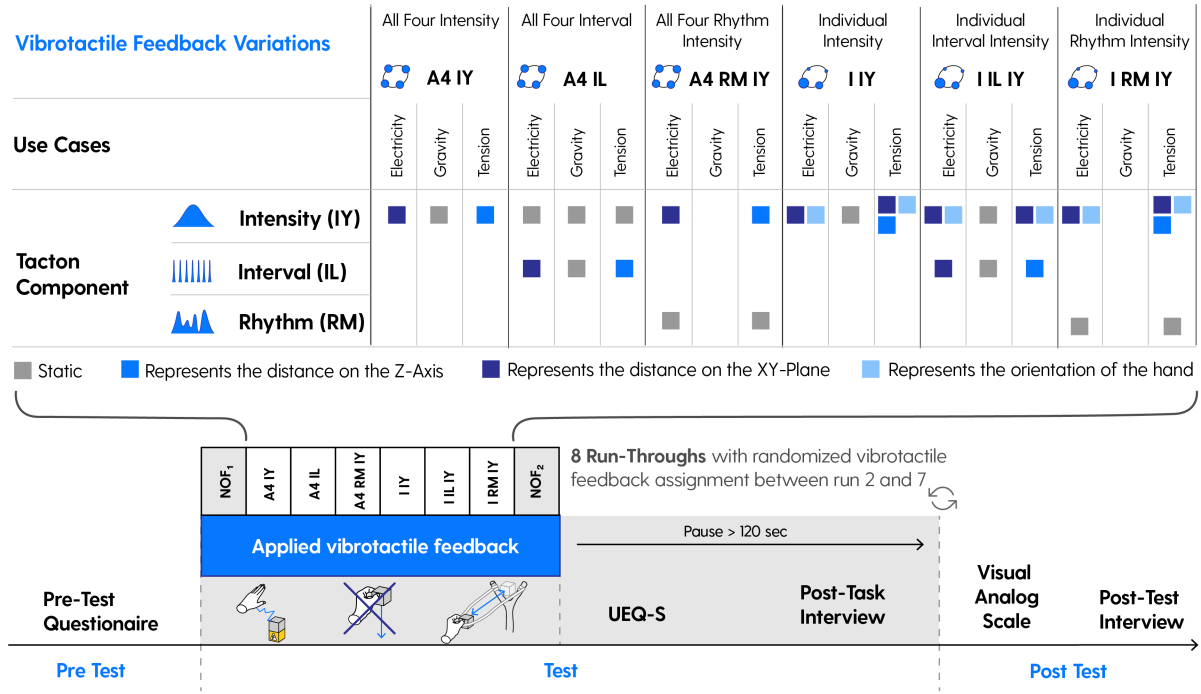


Figure 4: Study procedure. Each participant had to complete a task consisting of Electricity and Tension subtasks eight times in total with randomized vibrotactile feedback variants.

sec, depending on the user's hand distance to a reference point.

- **Rhythm (RM):** The rhythmic patterns with a duration of ca. 2 sec. were generated individually for each property using Apple Haptic and Audio Pattern (AHAP) files [4], which loop as long as the user interacts with the object.

4 USER STUDY

The different vibrotactile patterns were evaluated in a user study aiming at identifying the general acceptance/quality of the wearable and the usability of the different created vibration patterns. Special attention was paid to user experience (UX) and, thus, to the pragmatic and hedonic effect of the vibrotactile feedback on the interaction (as previous studies have already reported on the increased information content of vibrotactile messages and their spatial layouts [14]).

4.1 Participants

As previous research has shown that sensitivity to vibration decreases with age [6, 7], subjects aged between 20 and 40 were considered for our study. In total, the study sample consisted of $n=18$ participants (8 female, 10 male) with an average age of $M = 27.16$ years ($SD = 5.67$). All participants were right-handed and based on self-estimation the experience with mid-air gesture interactions ranged from “none” (2), “low” (8), “average” (3), “high” (4) to “expert” (1).

4.2 Task and Procedure

To evaluate the vibrotactile feedback variants, a task-based approach was chosen in which the subjects experience the simulation of forces based on a concrete application example. We have combined the use cases electricity, gravity, and tension as described in Figure 3 into one comprehensive task: 1. Approaching an **electrical** charged object, 2. Placing the object into a slingshot by lifting it and 3. Shooting the object against a wall by **tensioning** the slingshot. However, we decided not to apply vibrotactile feedback to gravity, as it would be present all the time and potentially lead to overstimulation. In addition, only a few vibrotactile stimuli would be suitable to represent constant force. Subjects were asked to repeat this task eight times, starting and ending the series with the “no feedback” (NOF₁, NOF₂) condition as baseline. The six different feedback variants (A4 IY, A4 IL, A4 RM IY, I IY, I IL IY, I RM IY) were randomly assigned (Latin Square) to the other runs (see Figure 4). In tasks 2-7, vibrotactile feedback is applied to each of the two use cases electricity and tension. To prevent arm fatigue which could harm subsequent runs, we have introduced a resting phase between each task. After each task, the interaction with the actual vibrotactile pattern was evaluated using the short version of the user experience questionnaire (UEQ-S) [21] and a short post-task interview. After completing all eight tasks and experiencing all six different vibration patterns, the subjects were asked to rate each again using a visual analog scale (VAS). The rating is provided with a marked position on a two endpoint-anchored line of 10 cm – typically converted to a 101-point scale [9]. Before participants were finally debriefed, we conducted a post-test interview. For matter

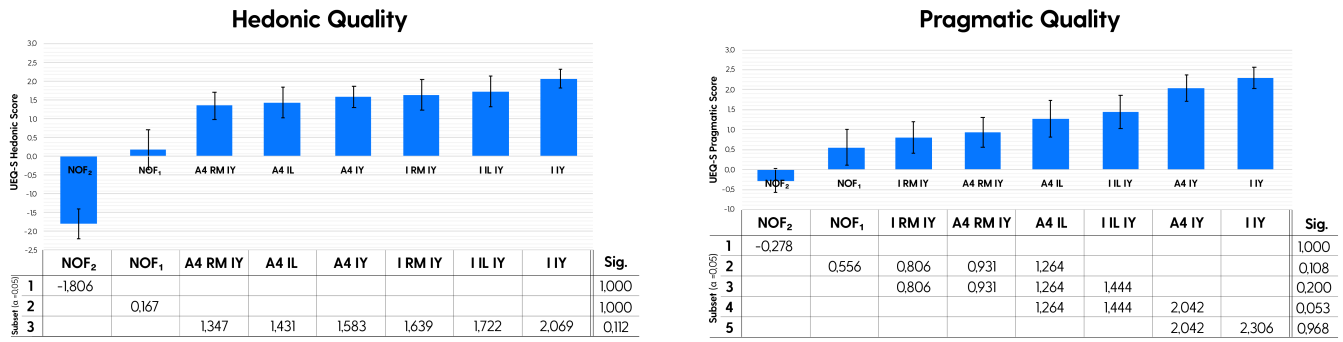


Figure 5: Descriptive statistics: Mean values (CI 95%) of UEQ-S for the six vibrotactile feedback variants and baseline (no feedback, NOF). All variants are presented in homogeneous subsets and sorted in ascending order by their mean values.

of space, the qualitative results from the post-task and post-test interviews are not reported here.

5 RESULTS

For the data from the pragmatic (N=144) and hedonic (N=144) subscale of the UEQ-S as well as the VAS (N=126), residuals were formed and tested for normal distribution (QQ plots). Furthermore, homoscedasticity was verified (Levene's test $p > 0.05$). Based on the normal distribution of all collected data, ANOVAs were performed regarding the pragmatic and hedonic subscales of the feedback variants as well as the results of the VAS. Subsequently, the variants were compared pairwise with a Tukey-HSD post-hoc test and grouped into homogeneous subsets after non-significant differences. All data collected were tested for significance with $p < 0.05$.

5.1 General Effects of the Vibrotactile Feedback and User Experience (UX)

As expected, the interaction with applied vibrotactile feedback was rated better than no applied feedback. In particular, by investigating the influence of vibrotactile feedback variants on evaluating the interaction without feedback afterwards (NOF₂) compared to before (NOF₁), it was found that NOF₂ was assigned a significantly worse pragmatic quality than NOF₁ ($p = 0.029$), see Figure 5. This is also evident concerning the hedonic quality ($p = 0.000$).

The ANOVA of the hedonic subscale of the UEQ-S of each variant revealed significant differences between the variants ($F = 47.365$, $p = 0.000$). The subsequent Tukey-HSD post-hoc tests revealed three homogeneous subsets in which no significant differences occur. NOF₂ ($M = -1.806$, $SD = 0.807$) forms subset 1 with the significantly lowest hedonic quality followed by subset 2 consisting of NOF₁ ($M = 0.1667$, $SD = 1.067$). Subset 3 contains A4 RM IY ($M = 1.347$, $SD = 0.718$), A4 IL ($M = 1.431$, $SD = 0.813$), A4 IY ($M = 1.583$, $SD = 0.569$), I RM IY ($M = 1.639$, $SD = 0.819$), I IL IY ($M = 1.722$, $SD = 0.826$), I IY ($M = 2.069$, $SD = 0.506$) and forms the group with the significantly best hedonic quality compared to subset 1 and 2. Within subset 3 there are no significant differences ($p = 0.112$).

As for the pragmatic subscale of the UEQ-S, the ANOVA showed significant differences for each variant ($F = 21.101$, $p = 0.000$). The Tukey-HSD post-hoc test yields five homogeneous subsets in

which no significant differences occur. NOF₂ ($M = -0.278$, $SD = 0.606$) forms subset 1 with the lowest significant hedonic quality. The variants NOF₁ ($M = -0.278$, $SD = 0.606$), I RM IY ($M = 0.806$, $SD = 0.793$), A4 RM IY ($M = 0.931$, $SD = 0.752$) and A4 IL ($M = 1.264$, $SD = 0.925$) form Subset 2 ($p = 0.108$). The feedback variants I RM IY, A4 RM IY, A4 IL and I IL IY ($M = 1.444$, $SD = 0.851$) in Subset 3 show no significant differences ($p = 0.200$). Subset 4 ($p = 0.053$) is represented by A4 IL, I IL IY and A4 IY. ($M = 2.042$, $SD = 0.655$). A4 IY and I IY ($M = 2.306$, $SD = 0.532$) form subset 5 ($p = 0.968$) with the comparatively highest pragmatic quality. Based on the homogeneous subsets, significant differences can be identified.

5.2 General Rating

The ANOVA of the results of the VAS of each variant showed significant differences ($F = 28.324$, $p = 0.000$). The Tukey-HSD post-hoc test yields four homogeneous subsets grouped by non-significant differences. NOF ($M = 24.833$, $SD = 12.962$) in subset 1 ($p = 1,000$) was significantly worst rated of all variants. A4 RM IY ($M = 50.167$, $SD = 15.421$), A4 IL ($M = 54.556$, $SD = 16.336$), I RM IY ($M = 59.056$, $SD = 16.264$) and I IL IY ($M = 64.000$, $SD = 15.856$) in subset 2 ($p = 0.059$) show no significant differences. A4 IL, I RM IY, I IL IY and A4 IY ($M = 64.667$, $SD = 11.449$) form subset 3 ($p = 0.143$). I IY ($M = 82.444$, $SD = 9.024$), only in subset 4 ($p = 1.000$), was evaluated as significantly best vibration pattern.

6 DISCUSSION

The significant differences between NOF₁ and NOF₂ suggest that the feedback of the wearable adds another dimension to the UX of virtual object interactions as the subjects have negatively changed their attitude towards the baseline interaction after the demonstration of the vibrotactile feedback variants. The extreme decrease in the hedonic quality of NOF₂ indicates that the vibrotactile feedback shaped the interaction with the objects in a novel, more exciting and interesting way as forces and phenomena were perceived as such through congruent visual and proprioceptive feedback as stated by the subjects. In addition, participants reported improved transparency of interaction with objects, suggesting that such effects can also be used as a proxy of classical mouse events for gesture interactions (i.e., phenomena such as hover effects or the representation of forces for drag-and-drop tasks). The clear pragmatic

tendency towards the continuously constant patterns A4 IY and I IY can be justified on their uninterrupted feedback, providing timely accurate statements about the user's hand's spatial position. Furthermore, the subjects reported on an improved orientation and interaction precision as visual-spatial reference points were paired with relative vibrotactile cues. Due to the inter-stimulus interval's duration, discontinuous patterns A4 IL and I IL IY have a period of time during which the user does not receive any feedback despite possible changes in the force magnitude or distance but rather perceives this delayed. However, absolute distances could still be better determined. The rhythmic patterns A4 RM IY and I RM IY partly led to confusion as they dynamically change their base profile on their own and not to the user's actions, justifying their rather low pragmatic quality. Rhythmic patterns thus seem to be more suitable for static abstract messages with constant baseline intensity.

The tendency to favor the three-dimensional presentation of the object properties suggests an increased number of vibration motors can enhance virtual objects' perception through spatial cues. However, as the significant differences between I RM IY and I IY show, a vibrotactile feedback's success is not determined by a single factor such as a vibration parameter or spatial arrangement in general, but by the correct composition and the coordination of these factors.

7 CONCLUSION

We presented a wrist-worn wearable for mid-air gesture interactions with virtual objects that transforms undetectable physical forces, phenomena, and spatial relation in 3D space into a haptic experience. We utilized a one-dimensional actuator array around the wrist, accompanied by a set of vibration patterns that change dynamically based on the user's action. Based on a virtual slingshot game, we evaluated the general feasibility and acceptance/UX of the prototype and its vibrotactile messages within a user study (n=18). Results suggest several strengths and weaknesses for each individual vibration pattern. Simple patterns of constant vibration mediated synchronously by actuators and change based on user input can already cause a significantly improved mid-air gesture interaction and evoke strong associations of the seen with congruent behavior. Since there are myriad of smartwatch users, our findings demonstrate the enormous unused potential of those devices in the context of ubiquitous computing, which can counteract the major drawback of mid-air gesture interactions' lack in haptics. That three-dimensional representations of vibrations around the wrist achieve a slightly improved UX is a positive side-effect, which nevertheless presents the capability for advancing smartwatches. It encourages thoughts and ideas of developing them further in the future, as the development of smart materials or textiles could also enable tactile sensations [8, 15].

This work has some **limitations**, including small sample size, young age of participants, subjective measurements only, and right-handedness, which will be resolved in further studies. Furthermore, since we are only testing three properties, the results cannot be generalized. In particular, in **future work** the different variants could be examined in more detail regarding the performance increase of the interaction through objective measurements. Another

improvement would be to replace ERMs with linear resonant actuators (LRAs). We assume that this would allow vibrotactile messages to be designed more precisely and with fewer design constraints. This should also have a positive effect on the vibration patterns tested.

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We applied the SDC approach to arrange the sequence of authors.

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